

ArkhamExpress

Task 1 : Research

1. What is a workflow and what is a workflow management system?
 - A workflow is a structured sequence of activities that transforms and input into an outcome, defining what work must be done, who performs it, in what order and what happens when it succeeds, fails or if it requires escalation. A workflow management system is a software that stores workflow definitions and uses them to create and manage many running instances of that process, coordinating whichever of work items, participants, routing, approvals, events, integration, history and reporting the process. **JIRA** is a practical example, letting teams define customizable workflows suited for their own needs. (Atlassian, n.d.)
2. How are activities or work items represented?
 - Across multiple workflow managing platforms, activities and or work is represented as a table that looks like an Excel spreadsheet. In the table, common categories like a summary of the task, the status (TO-DO), a due date, the priority of the task and comments that may have been left by other teammates. These are the most important things a user needs to know about the task.
3. How are work items assigned and routed?
 - Tasks are directly assigned to users based on their user ID (Oracle, n.d.). Tasks can be shared to a queue or to a specific role rather than a single person, any member that has that specific role can either claim or be assigned to a task. Tasks can also be assigned conditionally, factors like priority, budget, amount, department or even geo-location can decide who receives the task. They are usually routed in the following 3 ways, **sequential** - here tasks move in a strict linear chain where completion of one task immediately triggers a route to the next tasks in line, **parallel** – a single step branches out into multiple simultaneous tasks assigned to different people or departments at the same time, **conditional** – the path of workflow changes based on an outcome or decision and finally, **escalation & fallback** – if a task expires of the designated user isn't available, the workflow manager automatically re-routes or escalates the item to a supervisor or an alternative handler.
4. What is the difference between a workflow definition and a running workflow instance?
 - A workflow definition is the reusable template for a process, created once and describing its stages and legal transitions, while a workflow instance

is one live, independent execution of that template, with its own participants, data and progress. Editing or deleting a running instance has no effect on the definition it was created from, and the same definition can spawn many instances running at completely different points in the process simultaneously. In Camunda, this is enforced directly: a deployed process definition is immutable once instances are running against it, and each instance tracks its own variables and current state independently of every other instance (Camunda, n.d.-b). This is the foundational separation TIVIDY is built around, so it needs distinct definition and instance classes from the very start rather than a single class that tries to do both jobs.

5. How are approvals, rejection and escalation handled?

- Approval steps require one or more designated participants to review and either approve or reject a work item before it proceeds; a rejection typically routes the item back to the submitter or a prior stage rather than stopping the workflow outright. In ServiceNow, approvals can require a single approver, a unanimous group, or majority voting, and the system automatically escalates to a backup approver if a deadline passes without a decision (ServiceNow, n.d.). This pattern repeats across most of the platforms researched, so TIVIDY needs configurable approval logic rather than one hardcoded approve/reject check.

6. How are participants, roles and responsibilities represented?

- Participants are usually represented separately from the tasks they perform, as users, groups, or roles that work items get assigned to, rather than named individuals hardcoded into the process itself. Camunda models this with candidate groups and candidate users, so a task can be claimed by anyone holding a given role instead of needing to know in advance who specifically will handle it (Camunda, n.d.-a). Since responsibilities shift as organisations reorganise, representing participants abstractly by role needs to be a core part of TIVIDY's design rather than fixed inside individual tasks.

7. How are dependencies, events and failures handled?

- Dependencies are usually modelled as explicit links between tasks that control ordering, a task cannot start until its prerequisite finishes, while events represent things that happen during execution, a timer expiring, a message arriving, an error being thrown, that can start, interrupt or redirect a process. BPMN, used by engines like Camunda and jBPM, defines dedicated timer, error and message event types specifically so failures and triggers can be modelled directly rather than hidden inside code (OMG, 2013). This suggests TIVIDY needs events treated as first-class objects the engine can react to, not exceptions caught ad hoc.

8. How do workflow systems interact with external systems?

- Workflow systems typically expose integration points, often called connectors, service tasks or webhooks, that let a running process call out to an external system, sending an email, charging a payment, updating a separate database, partway through execution. Microsoft Power Automate works almost entirely around pre-built connectors to hundreds of third-party services, treating each integration as a swappable unit rather than special-cased code inside the workflow engine (Microsoft, n.d.). This is a strong signal that TIVIDY should isolate integration logic behind a common interface, since the specific external systems involved will vary per deployment.

9. How is workflow history or audit information recorded?

- Most systems log every state change, assignment and decision made during a workflow instance's life as an append-only, timestamped audit trail tied to whichever user or system caused it, so the process's history can be reconstructed later for compliance or debugging. SAP Business Workflow keeps a dedicated work item history log separate from the live process data, specifically so audits do not touch or risk corrupting the running instance (SAP, n.d.). This suggests TIVIDY should treat history as a record attached to each instance, decoupled from the instance's current working state.

10. How are running workflows monitored and reported on?

- Monitoring tools typically track running instances in aggregate, how many are active, how long tasks are taking, where bottlenecks are forming, usually through a separate dashboard or reporting layer rather than functionality built into the core workflow engine itself. Camunda Optimize ships as an entirely separate product from the Camunda engine for this exact reason, so reporting features can evolve independently of the core runtime (Camunda, n.d.-c). This reinforces that TIVIDY's monitoring and reporting responsibility should be a separate, pluggable layer that observes instances rather than logic embedded inside the work-item classes themselves.

11. What makes a workflow engine reusable across different processes?

- A workflow engine becomes reusable across different processes when the engine itself contains no process-specific logic, routing decisions and approval chains are supplied externally as configuration or pluggable rules rather than hardcoded into the engine's classes. jBPM and Camunda are both marketed as general-purpose engines precisely because the same runtime can execute entirely different process definitions without any change to the engine's source code (Red Hat, n.d.). The implication for TIVIDY is that genericity has to be a deliberate

design constraint, since any process-specific assumption baked into the engine breaks reusability.

12. Which information belongs to a workflow definition and which belongs to a running instance?

- A workflow definition holds everything that describes how a process is allowed to behave, its stages, legal transitions and rules, while a running instance holds only the data specific to one execution of that process, its current state, participants and progress. In Camunda, a BPMN process definition is deployed once and referenced by many separate process instances, each with its own variables and execution state, without any instance ever modifying the shared definition (Camunda, n.d.-b). This separation is consistent enough across the research that TIVIDY needs genuinely distinct class hierarchies for definitions and instances, not a single class trying to do both jobs.

13. How do real workflow systems model nested activities or sub-processes?

- Real workflow systems typically model a sub-process as a self-contained process definition that can be invoked from within a parent process, treated by the engine as a single step from the parent's perspective but internally made up of its own stages and transitions. BPMN explicitly supports call activities and embedded sub-processes for this reason, letting a large process be broken into smaller, independently testable pieces instead of one flat, unmanageable diagram (OMG, 2013). This points to TIVIDY needing a composite-style structure, where a single unit of work and a group of work can both be treated uniformly by the rest of the system.

14. Which decisions are normally configurable rather than hard-coded into the workflow engine?

- Decisions that tend to vary by organisation or change frequently, routing and assignment rules, approval chains, escalation timeouts, which integrations fire on which event, are the ones real systems keep configurable, while the core mechanics of moving a work item through valid states generally stay fixed inside the engine. Appian lets non-developers configure routing and approval rules through a visual designer rather than requiring a code change, specifically because those rules are expected to change far more often than the underlying engine (Appian, n.d.). This is the clearest signal from the research for where TIVIDY needs pluggable, pattern-based extensibility rather than hardcoded logic.

15. How are changes to workflow definitions handled while work is already in progress?

- Most mature workflow systems version their definitions rather than overwriting them, so a running instance keeps executing against the exact

version of the definition it started under, while new instances pick up the newest version going forward. Camunda explicitly supports multiple deployed versions of the same process definition running at once for this reason, since forcing an in-flight instance to suddenly follow a changed process could leave it in an invalid or undefined state (Camunda, n.d.-b). This means TIVIDY's definitions cannot be treated as mutable objects instances simply point to, versioning needs to be built in from the start.

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